PHINMA LIBRARY MANAGEMENT SYSTEM

System Overview | ITE298
PHINMA University of Iloilo
Instructor: Jayrold Tabalina

Prepared by: _____
Group/Section: _____
Date submitted: _____

1. Project Purpose

The Problem

Book records can be difficult to track manually. Library staff need to maintain book and student information and ensure that borrowing and returns update available copies.

Our Solution

The PHINMA Library Management System keeps key records in one application, supports book and student management, records borrowing and returns, and displays summaries on a dashboard.

Who It Is For

The intended users are library staff and authorized users who maintain catalog records and circulation transactions.

2. Architecture & Tech Stack

Typical flow: Browser → Apache/PHP → Database → Result displayed in the browser.

3. Key Modules

4. Data Flow: Borrow a Book

1. Select a student and book and enter the due date.

2. PHP checks the submitted information and book availability.

3. The database saves the borrowing record and updates available quantity.

4. The system displays the saved transaction and updated quantity.

Flowchart: Start → Select student/book → Validate student and availability → Save borrowing record and reduce quantity → Show result → End. If validation fails, show an error and do not save the loan.

5. Data Flow: Return a Book

1. Staff selects an active borrowing record.

2. PHP checks that the loan has not already been returned.

3. The database updates the return date/status and restores available quantity.

4. The system displays the completed return.

Flowchart: Start → Select active loan → Check status → Update return date/status and restore quantity → Show result → End. If the loan is not active, do not process the return again.

6. Communicating with Audiences

Technical Audience

“PHP validates the request, saves records using database queries, and updates availability within a transaction.”

Non-technical Audience

“The system records the loan and shows one fewer copy available until the book is returned.”

Use technical terms with instructors and developers. Use clear, everyday language with library staff and general users.

7. Demo Structure

Presentation tips: keep a steady pace, show the running system, explain technical terms clearly, and invite questions.

Feedback Method: Situation–Behavior–Impact (SBI)

Situation identifies when something happened; Behavior describes what the presenter did without judgment; Impact explains how it affected the audience.

Example: “During the borrowing demonstration (Situation), the presenter showed how the available quantity changed (Behavior), which helped the audience understand how the system tracks books (Impact).”

8. Self-Audit & Submission

Self-Audit Checklist

■ Database operations save the expected records.

■ Input validation rejects invalid values.

■ Borrowing and returning update quantities correctly.

■ A returned loan cannot be processed again through the normal workflow.

■ Error messages are clear and do not reveal sensitive details.

■ Login and protected-page access are tested.

■ The project works using the included SQL export.

Submission Package

• Project workspace containing PHP pages and CSS assets.

• SQL schema script: phinma_library.sql.

• System deployment guide.

• Technical documentation with group details and submission date.

Local Deployment Guide

Copy the project folder into the active XAMPP htdocs directory.

Start Apache and MySQL/MariaDB in XAMPP.

Open http://localhost/phpmyadmin and create/select phinma_library.

Import phinma_library.sql.

Check database settings in config/connection.php.

Open http://localhost/phinmalibrary/ (adjust if the folder name differs).

Test login, books, students, borrowing, and returning.

Note: Review sample records before sharing the SQL file. Do not publish real student details, credentials, or password hashes. Activity-log writing and role-specific permissions should be tested before claiming they are fully implemented.

9. Conclusion and Future Work

The system provides a foundation for managing books, students, borrowing, and returns through a browser-based application connected to a relational database. Possible future improvements include activity-log writing, stronger role-based permissions, additional reports, and more validation and security testing.

Component	Technology / Responsibility
Frontend	PHP-generated HTML and CSS pages shown in a browser.
Styling	assets/style.css controls page layout and appearance.
Backend	PHP handles sessions, input validation, and library operations.
Database access	MySQLi connection in config/connection.php.
Data storage	MySQL/MariaDB-compatible database named phinma_library.
Local environment	XAMPP provides Apache and MariaDB/MySQL for local testing.

Module	Description
Login & Logout	Checks account credentials and manages sessions.
Dashboard	Shows book quantities, available copies, students, and borrowing summaries.
Book Management	View, add, edit, and delete book records.
Student Management	View, add, edit, and delete student records.
Borrowing	Creates a borrowing record and reduces available quantity.
Returning	Updates the loan status/date and restores availability.
Admin Account Setup	Creates an account and stores a password hash.

Stage	Demonstration Activity
Intro	Explain the library problem and system purpose.
Feature	Show login, dashboard, book management, and student management.
Data Flow	Demonstrate borrowing and returning and explain the database changes.
Conclusion	Summarize current features, limitations, and possible improvements.